NATIONAL INSTITUTE OF TECHNOLOGY , ANDHRAPRADESH

Smart Microgrid Energy Forecasting & Optimization

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INTRODUCTION

The increasing use of renewable energy sources such as solar and wind introduces challenges because their power generation is highly variable and depends on weather conditions. At the same time, electricity demand also changes throughout the day, creating a mismatch between energy generation and consumption in microgrids. This project focuses on forecasting load demand along with renewable energy generation and using these predictions to determine an efficient energy management strategy for better utilization of available power sources.

PROBLEM OVERVIEW AND PROJECT GOALS

Renewable energy sources such as solar and wind are highly variable and depend on environmental conditions, making power generation uncertain.Electricity demand changes continuously, creating a mismatch between energy generation and consumption in microgrids.This mismatch can lead to inefficient utilization of available energy resources.The objective of this project is to forecast electrical load demand and renewable power generation.The predicted values are then used to support better energy management and improve the efficiency of microgrid operation.

PROPOSED SOLUTION

Develop an intelligent system to predict electrical load demand and renewable energy generation.Use historical microgrid telemetry data to learn patterns in energy consumption and generation.Forecast future load demand along with solar and wind power output.Utilize these predictions to support optimal energy management decisions.Allocate energy efficiently from available sources such as solar, wind, and the grid.Improve overall microgrid efficiency, reliability, and utilization of renewable energy resources.

DATASET DESCRIPTION

- The dataset contains time-series microgrid data used for analyzing electricity demand and renewable energy generation.
- It includes temporal features such as Time, Season, and Day_of_the_week to capture daily and seasonal patterns.
- Weather-related features include DHI, DNI, GHI, Wind_speed, Humidity, and Temperature, which influence renewable power production.
- PV_production and Wind_production represent energy generated from solar and wind sources.
- Electric_demand indicates the total electricity demand of the system.
- These features help model the relationship between weather conditions, renewable generation, and load demand.

DATA PREPROCESSING

- The dataset is first cleaned by handling missing values and removing invalid records to ensure data quality.
- Time-based features are processed to extract useful patterns related to daily and seasonal variations.
- Feature engineering techniques such as lag features and cyclical time encoding are applied to capture temporal dependencies in the data.
- All input features are normalized using a scaling technique to ensure stable and efficient model training.
- The processed data is then converted into time-series sequences so that the model can learn patterns from historical observations.
- Finally, the dataset is divided into training, validation, and testing sets for proper model evaluation.

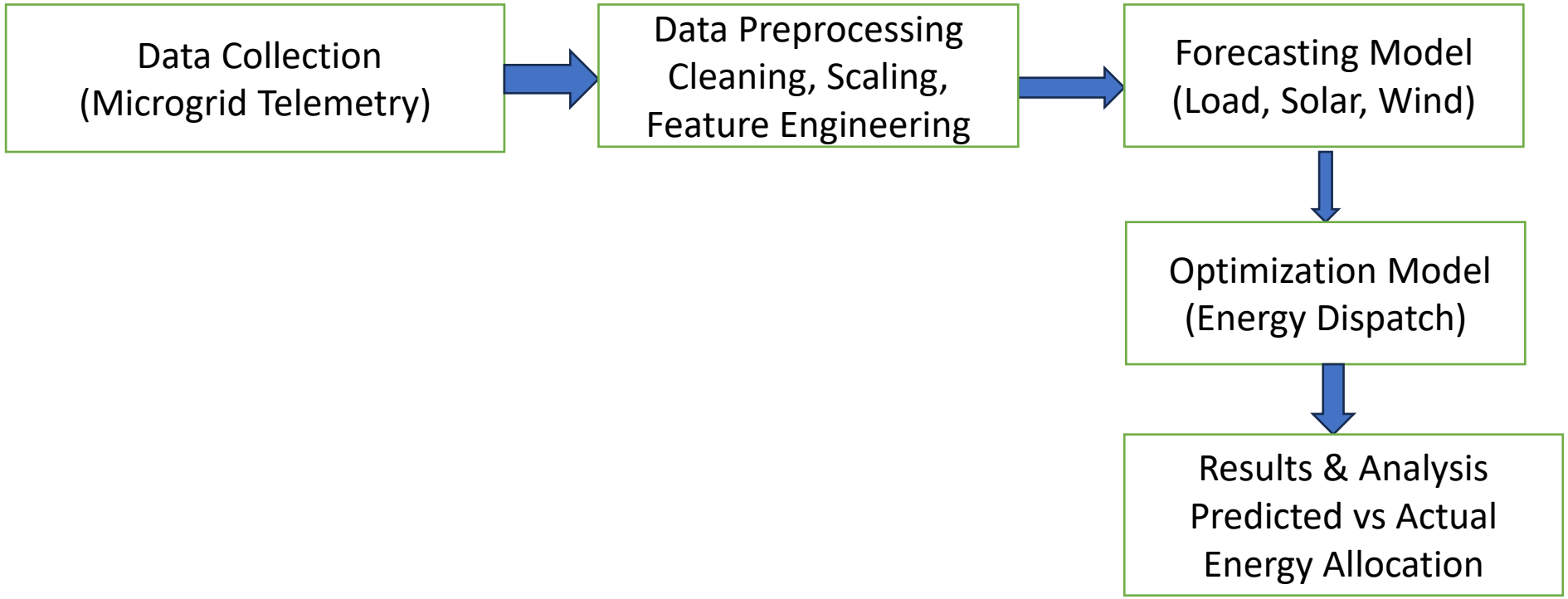
MODELS USED

- CNN–LSTM Model: Used for forecasting future electric demand, solar PV output, and wind power generation from historical time-series data.
- CNN extracts important patterns from the input features, while LSTM captures temporal dependencies in the data.
- Reinforcement Learning: Used for optimal energy dispatch, deciding how much power should be supplied from solar, wind, and grid to meet the predicted demand efficiently.

OPTIMIZATION METHODS

- Optimization Method
- Uses forecasted load, solar, and wind values to plan energy dispatch.
- Determines the optimal mix from solar, wind, battery, and grid to meet demand.
- Considers system limits and aims to minimize cost while maximizing renewable use.
- Produces an efficient energy dispatch schedule for the microgrid.

PROJECT WORKFLOW



CONCLUSION

This project successfully integrates deep learning and reinforcement learning for smart microgrid management. The CNN-LSTM model accurately forecasts solar, wind, and load demand, while the PPO-based RL agent optimally allocates energy resources. The system reduces grid dependency, improves renewable utilization, and ensures efficient energy balancing, making it a promising solution for sustainable and intelligent power systems.”